

Freezer Instructi On Manual

Dear Customer:

Thank you for your trust and support of our company, choose our production of quick-freezing cabinet. In the use of this product before you carefully read this manual, and strictly in accordance with the instructions for operation, if there is any uncertainty can be contacted with our company.

I. This series of products has the following characteristics:

1. Adopt imported compressor, low noise, high energy efficiency.
2. Adopt hot gas defrosting, safer and save electricity.
3. Adopting microcomputer temperature control system, can control the temperature within the permissible range.
4. Cooling and freezing quickly so that the food can be quickly frozen in a short time.

II. Security considerations:

1. Please read this "Safety Precautions" carefully before use so that it can be used correctly.
2. The precautions described herein are intended to enable you to

Maintenance records:

Repair Date:

Fault Cause Detection Results:

Maintenance Program:

Maintenance signature:

Maintenance records:

Repair Date:
Fault Cause Detection Results:
Maintenance Program:
Maintenance signature:

use the Quick Freezer safely and to prevent the occurrence of any hazard to your safety or the safety of others. The safety precautions are categorized as "WARNING" and "CAUTION" depending on the contents. Please be sure to observe these safety precautions.

III. Warnings:

1. Must use a separate three-hole socket and with the appropriate meter and fuse (if 380v to be equipped with an independent air switch), do not share with other electrical appliances with multiple sockets, in order to avoid danger.

2. The grounding end of the socket should have a reliable grounding wire, in addition to the grounding wire, should also be installed leakage protector.

3. The plug should be correctly inserted and pulled, inserted with force, so that the plug and jack contact is good, unplugged should be pinched plug, can not pull the power cord, in order to prevent the plug loose or broken wire.

4. Do not damage the insulation layer of the power cord, not heavy pressure on the wire, not to replace or lengthen the power cord at will, if the power cord is damaged, in order to avoid the danger of removing must be replaced by specialized personnel.

5. Do not use flammable sprays near the freezer/refrigerator,

otherwise there is a possibility of fire or explosion.

6. When the freezer / freezer around the combustible gases such as leaking gas, liquefied petroleum gas, etc. exists, the first to open the window to remove the surrounding combustible gases, do not pull the plug or turn off the thermostat and other switches, in order to avoid the generation of sparks caused by the explosion.

7. Please unplug the unit from the power supply during maintenance to prevent electrical shock.

8. Do not unplug or plug in the power supply with wet hands to prevent electric shock.

9. If there is any abnormality, cut off the power immediately and stop the operation. Do not turn on the power until the abnormality is removed, as abnormal operation may cause danger.

IV. Cautions:

1. The power supply requires 220V (380v), 50Hz, if the voltage is too high or too low, the compressor will be easily damaged.

2. Please do not touch the evaporator surface of the inner wall of the freezer/cooler with wet hands to avoid frostbite.

3. This freezer/cooler is not for refrigeration, only for fast freezing of food, the temperature of food should not be higher than 25 degrees.

4. Do not touch the exposed piping and wiring, which may cause

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burns or electric shock.

5. Do not put your hands inside the machine, there are pipes in high temperature inside the machine to prevent burns.

6. Remove the plug from the power outlet when moving the freezer/cooler.

7. When the freezer/cooler is not in use for a long time, please unplug it from the socket for safety reasons.

8. Non-professionals are not allowed to disassemble, repair or modify.

9. Whether moving or transporting, should avoid collision or severe vibration, freezer / freezer tilt angle should not exceed 45 degrees, in order to avoid damage to the refrigeration system.

10. The freezer/cooler should be placed in a dry place, if it is kept in a wet or watery place for a long time, the case will rust and affect the electrical insulation strength. Do not use it outdoors or in a rainy place.

11. Don't put the freezer/cooler in direct sunlight or near the heat source, otherwise it will affect the insulation layer of the freezer and the life of the sheet metal.

12. In order to make the quick-freezing cabinet / freezer better ventilation and heat dissipation, quick-freezing cabinet should be left around at least a certain amount of space, in order to facilitate ventilation and heat dissipation, reduce energy consumption.

13. placed on the ground should be flat, solid. And keep the level, so that the display cabinets placed smoothly, out of safety needs, and can make the compressor work smoothly, reduce vibration and noise.

V. First-time use:

1. First remove the packaging of the freezer/refrigerator, place the machine on the level ground, dry and ventilated surroundings, it is recommended that the left and right sides of the reserved more than 1 meter, the back of the reserved distance of more than 60cm, in particular, can not be blocked on the right side of the air outlet, if blocked, it will affect the heat dissipation of the entire machine, and ultimately affect the refrigeration effect.

2. Please carry out a thorough internal cleaning and sterilization.

3. After standing for 12 to 24 hours, plug the power supply into a 220V(380V)/50HZ power supply with a safety ground.

4. Turn on the machine control switch, after a delay of 3 minutes to enter the cooling state.

5. Turn on the power for trial operation, because the freezer cabinet is hot, the initial cooling will be slow. When the cabinet temperature down to -40 degrees or so when you can place items, place items faster to try to avoid a greater loss of cold.

6. Each tray is recommended to place 2-4 kilograms of frozen

Equipment maintenance service warranty card

Dealer Information	User information
Dealers:	User Name:
Contact Person:	Equipment Model:
Phone/Fax:	Date of purchase:
Cell phones:	User contact information:
Equipment maintenance service warranty card (copy)	
User Name:	
Contact phone number:	
Product Model:	
Address:	
Factory number:	
Date of purchase:	

amount of fog out of the wind, which is a normal phenomenon.

※ If there is any problem with the quick-freezer/refrigerator, please check it according to this manual first, and if you still can't solve it, please contact our after-sales service department in time.

x. important statements:

Use of this product must be connected to a reliable safety ground.

products, because the frozen products release heat and refrigeration is proportional to the amount of frozen products, do not place too much frozen products, wanting to go faster is not enough.

7. Keep the cold air circulation in the cabinet unimpeded, do not block the air supply port.

8. The compressor of the freezer cabinet is a 3-minute delay protection, when the compressor starts to start, the freezer cabinet starts to enter the refrigeration state, start refrigeration.

9. The product will be automatically defrosted once every 4 hours of operation after power on, when defrosting, the cabinet will be shut down for 20 minutes, the temperature inside the cabinet will rise, which is a normal phenomenon, after defrosting and will restart refrigeration.

10. No load under the premise of the cabinet temperature up to -45 °C or -60 °C (about 40-60 minutes), into the product, due to the frozen product is different, frozen weight is different, the release of its own heat, the time will be different, please customers in accordance with the actual production, calculate the time required for quick-freezing, conventional products can be quickly brought to our factory to test the product consult online customer service. (Customer service data is for reference only, subject to actual production).

11. As the quick-freezing cabinet belongs to the low-temperature freezer, the cooling conditions of the equipment, placed items will lead

to differences in the effect of the equipment to produce quick-freezing, the operator must read the operating instructions in order to start operating the equipment.

12. As frozen food itself with heat, frozen food in the filling process, to determine whether the frozen food is frozen, according to the specific circumstances of the specific analysis, can not simply determine whether the cabinet temperature reached -45 °C or -60 °C.

13. Due to different frozen foods, the method of quick-freezing is also different. We do not promise that certain foods must be frozen within the specified period of time, and we are not responsible for any costs or losses incurred as a result of quick-freezing foods. Please adjust according to the actual production situation.

14. Please note that the capacity of this product is relatively small compared to a cold storage, so the quick-freezing can not be compared to a large-capacity cold storage.

5. Noisy:

5.1 Is the ground level and firm?

5.2 Is the freezer/refrigerator placed in a smooth position?

IX. The following phenomena are not malfunctions:

1. When the environment is too humid, sometimes there will be condensation on the shell of the freezer/refrigerator, dry it with a cloth.

2. Quick-freezing cabinet / freezer will sometimes have a slight sound of running water, this is the refrigerant in the pipeline flow of sound, is a normal phenomenon.

3 freezer / freezer work, the compressor, condenser will be hot, these are normal phenomena.

4. When a deposit more high-temperature items, freezer / freezer cooling time is longer is a normal phenomenon.

5. Higher ambient temperatures in summer, the freezer / freezer need to work for a longer period of time, in order to reduce the temperature of the required food to freeze quickly, this is a normal phenomenon.

6. Every 4 hours of operation will automatically enter the defrost state, at this time the temperature will rise slightly, there is a small

and whether the power cord is damaged?

2. Completely uncooled:

2.1 Is there a power failure?

2.2 Is the power plug unplugged in the power outlet?

2.3 Is the fuse disconnected?

2.4 Is the earth leakage protector working? (Where an earth leakage protector is installed)

3. Poor cooling:

3.1 Check whether the condenser is clogged with dirt, whether the ventilation is good, otherwise clean the condenser to improve ventilation?

3.2 Check whether the evaporator is too much frost, too much can be taken manually defrost or stop defrost?

3.3 Is the thermostat parameter setting appropriate?

3.4 Is the freezer/refrigerator under direct sunlight or too close to the heat source?

4. No downtime:

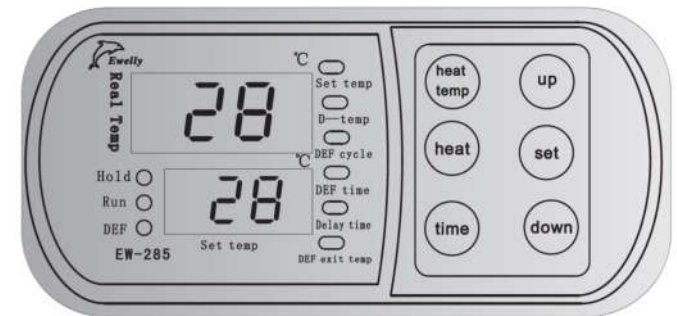
4.1 Is the ambient temperature too high?

4.2 Are the thermostat parameters set appropriately?

VI. Key Functions and Setting Methods:

6.1 -45°C models (285 Panel)

6.1.1 Panel Schematic:



6.1.2 Operational Profile:

Parameter setting: Press the function key until the set temperature light is on, every time you press the function key, you will enter the next menu setting (can be operated cyclically). After entering the menu, press ▲ or ▼ key to modify the parameters of the program.

6.1.3 Table of program parameters:

Manipulate	demonstrate	Setting range	Factory setting	
Hold [set] key for 6	[Set temp] light on	Setting range:-45°C~50°C	-45°C	Press [set] key to display the

seconds				original
Press the [set] key again	[D-temp] light on	Setting range:1°C~15°C	3°C	setting value, then press ▲ or ▼ key to
Press the [set] key again	[DEF cycle] light on	Setting range:1~99 hours	6 hour	change the setting value, and finally the
Press the [set] key again	[DEF time] light on	Setting range:1~45 minutes	10 minute	temperature control will run
Press the [set] key again	[Delay time] light on	Setting range:1~5 minutes	3 minute	automatically according to the new
Press the [set] key again	[DEF exit temp] light on	Setting range:5°C~35°C	5°C	parameters.
Hold [heat] key for 6 seconds	[DEF] light on	Controller first into the frost function, after the completion of the frost automatically enter the original set program work		
Press the [heat	None indicator	Display temperature of the defrosting sensor	Larger digital displays	

3. equipment after the lower left drain (if applicable), should be regularly drained or connected to the sewer.

4. It is recommended that users turn off the machine when the food is not frozen, which will consume less energy.

5. equipment should be used 2 months to clean up the condenser dust once, clean up should cut off the wires, will be removed after the ventilation plate, with a soft brush will be cleaned up the dust on the condenser, back to the ventilation plate can be installed, do not use tap water to rinse or use sharp tools to clean up, so as not to create a leakage or leakage of refrigerant.

6. After the maintenance is completed, please carry out the following checks to ensure safety.

7. Whether the power cord plug is completely inserted in the power outlet.

8. Whether the power plug has abnormal heat phenomenon.

9. Whether the power cord has cracks or scratches.

VIII. Anomaly check:

1. Not energized:

1.1 Check whether the power supply is energized, whether the power cord is plugged in properly, whether the operation switch is off,

		contactor.
Compressor overfrequency	Differential return temperature value is too low or pressure protector setting problem.	Readjust to increase return temperature value; adjust pressure protector setting.
Display E1, E2	Sensor line is shorted or open.	Check that the sensor wiring is connected to the connector.

※ Temperature adjustment must be adjusted according to the actual situation and the desired temperature, each adjustment needs to run for a period of time to stabilize.

VII. Maintenance and recommendations:

1. The equipment belongs to low-temperature freezer/refrigerator, can not be used as a storage cabinet. If 24 hours of use, the temperature does not go down, the evaporator is too thick frost, please stop 10 ~ 12 hours, or manual defrosting. Do not run for 7 consecutive days without interruption, so as not to strain the refrigeration system to your losses.

2. The devices on the equipment are fragile parts, should avoid collision, scrubbing should be careful when wiping to avoid damage.

temp] key	light is bright		
Press the [time] key	None indicator light is bright	It displays the remaining time in the run cycle during refrigeration, while displays defrosting remaining time when it defrosts	Larger digital means hours ,Smaller digital means minutes

6.1.4 Functional description:

Refrigeration: when the current temperature $\geq$ set temperature + return temperature (D) and delayed the set "delay time", (relay is engaged) the compressor starts, the operation indicator light is on, the thermostat indicator light off.

Refrigeration stops: when the current temperature $\leq$ set temperature, the refrigeration relay is disconnected, the operation lamp goes out. Constant temperature light is on.

Defrost start: after the controller runs the set "operation cycle" time after power on or defrost and the defrost sensor temperature $<$ defrost exit temperature, it will enter into defrost, the defrost indicator light will be on, the constant temperature light will be off, the refrigeration relay will be disconnected and the operation light will be off.

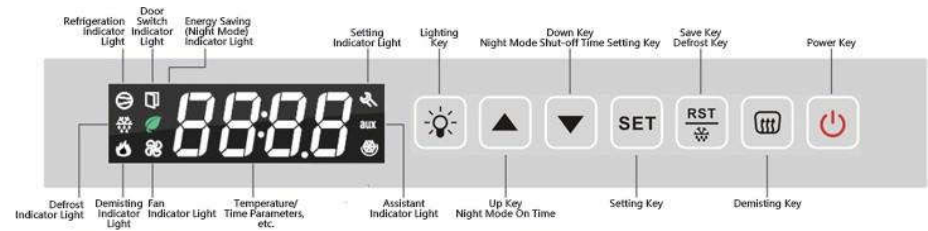
Defrost stop: when the frost sensor temperature $\geq$ frost exit

temperature or reached the set "frost time" to exit the frost, (frost has a default 2-minute drip time delay), frost indicator light off, thermostat light on. The controller resumes normal operation and starts the next frost cycle.

Note: When pressing the "forced frost" button, such as frost sensor temperature ≥ frost exit temperature, the system can not enter the frost state, to maintain the current operating state unchanged.

6.2 -45°C models （ 135 Panel ）

6.2.1 Panel Schematic:



6.2.2 Working conditions:

Operating voltage: 220VAC ±10%, 50/60Hz;

Compressor: 20A/220VAC; Defrosting: 10A/220VAC; Fan: 10A/220VAC;

Lighting: 10A/220VAC; Defogging: 10A/220VAC.

	the library temperature.	
The display temperature is unstable or has a false action	Sensor line interference; poor wiring; bundled up with its other power lines; line damage.	Separate the sensor from the power cord; change the shielded wire; check that the terminals are tightened.
Large error between the actual temperature value and the measured value of the thermostat	Incorrect installation position of each circuit sensor; too long wiring, high line resistance; poor wiring contact; sensor damage.	Each sensor must be installed in the correct position, increase the cross-section of the extension cord; ensure that the wiring is sealed, waterproof and moisture-proof; replace the sensor.
Reaching temperature without stopping	Incorrect installation of the sensor does not measure the correct value; compressor contactor failure.	Check that the sensor is accurately measuring the temperature; replace the compressor

		control with the dealer.
Displayed but not running	Set temperature higher than current temperature; external circuit protection.	Reset the required temperature control value; check the cause of external faults; reset the work after troubleshooting.
Frosting is not clean	Frost time is too short; frost exit temperature is too low.	Re-enter the menu to change the extended frost time; change to increase the frost exit temperature.
Frosting not working	Incorrect installation of the frost sensor; burnt out defroster; faulty control defroster wiring or components; set the frost exit temperature below	Recheck installation as shown; replace defroster; check wiring and components; reset to turn up frosting exit temperature.

Operating ambient temperature: -5°C to 60°C;

Operating relative humidity: 10% to 90% (no condensation);

Storage temperature: -25°C to 75°C;

6.2.3 Operation and display panel:

Refrigeration indicator light: Lights up when the compressor is operating, flashes when the compressor is delayed in starting, and remains off otherwise;

Defrosting indicator light: Lights up during defrosting, flashes when defrosting water drips, and remains off otherwise;

Defogging indicator light: Lights up during defogging and remains off otherwise;

Fan indicator light: Lights up when the evaporator fan is operating and remains off otherwise; Auxiliary indicator light: Lights up when the auxiliary relay is closed, and remains off at all other times;

Setting indicator light: Lights up when setting the shutdown temperature or other parameters, and remains off at all other times;

Energy-saving indicator light: Lights up during night mode, lights up when setting night mode parameters, and flashes during long holiday mode;

Door switch indicator light: Lights up when the door is open, and turns off when the door is closed

Lighting button: Turns the lights on or off (lights can still be controlled even when the unit is turned off);

Up button: Scrolls up through the menu in the menu interface and adjusts values upward in the parameter interface;

Down button: Scrolls down through the menu in the menu interface and adjusts values downward in the parameter interface;

Set button SET: Press and hold the Set button for 3 seconds to enter the parameter setting interface; switches between the menu and parameter interfaces;

Save Key/Defrost Key RST: View the evaporator sensor temperature (if present and functioning properly); save parameters and exit the parameter settings interface; hold this key for 3 seconds to start or stop manual defrosting;

Defog Key: Turn defogging on or off;

Power Key: In the powered-on state, hold for 6 seconds to shut down; in the powered-off state, hold for 6 seconds to power on.

6.2.4 Set operation (e.g., start-up temperature 2 ° C, shutdown temperature 8 ° C):

Set shutdown temperature: In the normal interface, press the “SET” button for more than 3 seconds until ‘st’ is displayed. Then press the “SET” button again. At this point, you can use the ▲▼ keys to adjust the shutdown temperature value to 2.0°C. The shutdown temperature value range is between C4 and C5.

Set startup temperature: Startup temperature = shutdown temperature + cooling hysteresis; In the normal interface, press and hold the “SET” button for more than 3 seconds until “st” is displayed. Then press the ▲ button again. The display will show “Po.” Press the “SET” button, and the display will show “0.” Use the ▲▼ buttons to adjust the display to 55, then press the ‘SET’ button. The display will show “C1.” Press the “SET” button to enter the parameter settings, then use the ▲▼ buttons to adjust to 6.0°C.

Note: Shutdown temperature = Set value; Startup temperature = Set value + C1 value;

(a) When the current temperature $\geq$ control temperature + return temperature (d) and delay the set "delay time (Pt)", the load relay is engaged, the operation indicator light, start refrigeration.

(b) when the current temperature $\leq$ control temperature, load relay disconnect, operation indicator light off, refrigeration stops.

2、Frosting control

(a) When the instrument runs the set operation cycle or long press " " key for 6 seconds, frost indicator light, load relay disconnect, after a delay of 1 minute, frost relay suction, load relay suction, start frost.

(b) when the frost running time $\geq$ frost time or frost state long press " " key 6 seconds, frost indicator light off, load relay disconnect, after a delay of 1 minute, frost relay disconnect, frost stop.

6.4 Common Troubleshooting:

fault phenomenon	Inspection of causes	cure
No display at power on	Whether the power supply is disconnected. Whether the temperature control is faulty.	Check the power supply wiring and replace the fuse; check the power supply side for input or replace the temperature

d	temperature return	0.5 ~ 15℃	8	With this parameter the target temperature can be controlled to stabilize in a certain set range.
Pt1	Compressor delay time	1 ~ 10	3	Prevents frequent start-up of the No. 1 compressor and protects the equipment.
Cb	Return Tube Temperature Correction	-15.0℃ ~ 15.0℃	0	When there is an error in the measured temperature of the return air pipe, the temperature can be corrected by this parameter.
Pt2 (Note 1)	Compressor No. 2 delay time	0 ~ 30	02	Prevents frequent start-up of the No. 2 compressor and protects the equipment.
Set2	Compressor No. 2 start-up temperature	-135 ~ 50℃	-5.0℃	Prevents the temperature setting from being too high.
F0	Frosting time	1 ~ 60 minutes	15	Frosting working time.
F1	Frost cycle (Note 2)	0 ~ 99 Hour	4	The amount of time the meter has run for the set amount of time before frosting is initiated.
F2	Drip time after frosting	0 ~ 30 minutes	2	
bF	Alarm Beep Setting	on/off	on	
bP	Touch key beep sound setting	on/off	on	

Note 1: Compressor No. 2 is running, when compressor No. 1 is running after the delay time of "Pt2" and when the "return pipe temperature" ≤ "Set2", compressor No. 2 stops running and stops at the same time as compressor No. 1. Stop operation at the same time with compressor No.1.

Note 2: When the defrost cycle is set to "0", it will not enter the defrost cycle, but it is still possible to manually force the defrost.

6.3.4 Functional description:

1、Refrigeration control

6.2.5 System Menu Settings:

In normal operating conditions, press and hold the "SET" button for more than 3 seconds until the temperature display window shows the parameter code "St," indicating entry into the system settings menu. Use the ▲ or ▼ keys to switch the parameter code item to Po; press 'SET' to display 0, then use the ▲ or ▼ keys to adjust the value to "55," and then click "SET" to enter the administrator menu, where you can adjust the basic parameters C1 to F6.

In the system menu settings state, pressing the "RST" key or remaining without any key presses for 30 seconds will save the modified parameter values and exit the system menu settings state. If an error occurs during parameter saving, the temperature display window will show "Err," and after 5 seconds, it will return to the normal display state.

6.2.6 Parameter table:

Parameters	Description	Set range	Factory default setting	Commentary
User menu				
Set	temperature setting value	Lower limit of temperature setting ~ Upper limit of temperature setting	4℃	
Po	Administrator Menu Password	-99 to 99 (password is 55 and cannot be changed)	55	(The factory menu password is -15.)
Administrator menu (password 55)				

C1	refrigeration hysteresis	0.5℃ ~ 9.9℃	4.0℃	4.0℃
C2	Minimum interval time between compressor starts	0 ~ 60 minute	5 minute	The minimum time interval between turning off and turning on the power again includes the startup delay when the power is turned on for the first time.
C4	Lower limit of temperature setting value	-40℃ to temperature setting value	-45℃	
C5	Upper limit of temperature setting value	Temperature setting value: 85℃	10℃	
d1	Evaporator Sensor Selection	0: Disabled 1: Enabled	1	
d3	Defrost cycle calculation method	0: Natural time 1: Cumulative cooling time	0	
d4	defrost cycle	0 to 90 hours 0: Defrosting prohibited	6	Time interval between two defrosting cycles (recorded automatically every hour)
d5	Defrosting and defrosting drip display mode (includes 15 minutes after defrosting drip)	0: Normal cabinet temperature 1: Displays dEF 2: Displays cabinet temperature when defrosting begins	1	Note: If the cabinet temperature reaches the temperature at the start of defrosting within 15 minutes after defrosting and dripping, the real-time cabinet temperature will be displayed.
d6	defrosting time	1 to 120 minutes	20 minutes	Defrosting duration
d7	defrost stop temperature	-40.0℃ to +50.0℃	10℃	When the defrost sensor temperature is higher than this set value, defrosting stops and is not allowed. Note: When d1=0, this parameter item is disabled.

parameter value (value flashing display), press the "↑" key or "↓" key to adjust the parameter value.

(c) Press the "Set" key to determine the modified parameter value, and return to display the parameter code.

(d) in the system setup state, press the "⏻" key to exit the menu settings, return to the normal operating state display.

(e) In the system setting state, no key operation within 6s, automatically return to the normal operation state display.

5、One-key restoration of preset parameters

(a) in the normal operating state, press the "↓" key for more than 6s, the display flashes, all the parameters are restored to the factory settings (please make sure to record the set parameters in advance).

6、Buzzer alarm silencing

(a) When the meter reports a fault, press any key to cancel the buzzer alarm sound, but the indicator does not go out until after the fault is lifted.

7. System parameters

parameters	descriptive	Setting range	Factory setting	marginal notes
LS	Control Temperature Limit	-151℃ ~ Control temperature-1	-61/-81	Limit only the temperature setting without alarm.
HS	Control Temperature Limit	Control Temperature +1 ~ 51℃	50	Limit only the temperature setting without alarm.

the standby state, displaying "----", all outputs are disconnected.

(b) In the standby state, press and hold the "⏻" key for 3s to return to the normal operation state.

2、Lock function

(a) the meter in the locked state, press the "↑" key 3s can be unlocked, accompanied by a buzzer sound, the meter in the absence of any key action 30s after the self-locking.

3、Temperature fast setting

(a) in normal operation, press the "Set" key to display the control temperature (value flashing display -20). b) Press the "↑" key or "↓" key to adjust the parameter value. c) Press the "↑" key to adjust the parameter value. d) Press the "↑" key to adjust the parameter value. e) Press the "↑" key to adjust the parameter value.

(c) Press the "↑" key or "↓" key without releasing, you can quickly adjust the parameter value (this is a general operation).

4、User parameter adjustment method

(a) Press the "Set" key for more than 6 seconds to the temperature display column shows the parameter code "LS", indicating that it has entered the system setup menu, press the "↑" key or "↓" key to switch the corresponding parameter code item. Press "↑" key or "↓" key to switch the corresponding parameter code item.

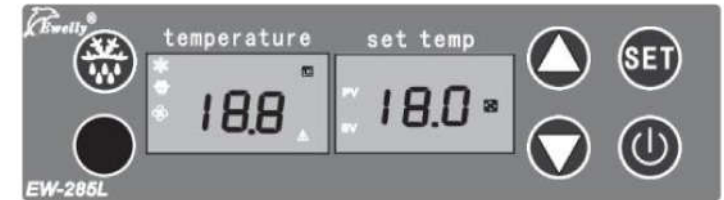
(b) Press the "Set" key to display the corresponding parameter code

d8	Defrosting drip time	0 to 120 minutes	3 minutes	Delay time after each defrosting cycle (during which the compressor is not allowed to operate)
d9	Defrosting type	0: Electric defrosting 1: Hot air defrosting 2: Air defrosting	0	Before using hot gas defrosting, the compressor should be shut down for 3 minutes, then restarted for defrosting to ensure the stability of the unit system. After hot gas defrosting, the compressor should be delayed for 2 minutes before restarting when conditions are met.
d10	Auxiliary heating compared to defrosting advance start time	0 to 60 minutes	15 minutes	0 Cancel delay
d11	Auxiliary heating compared to defrosting delay shutdown time	0 to 60 minutes	5 minutes	0 Cancel delay
d12	Auxiliary heating function	0: Off 1: On	0	
d13	Defrost start temperature	-50℃ ~ 30℃	30℃	Set to 30℃ Disable this feature
F1	Fan operating mode	-180 to -1: Fan starts 180 to 1 seconds before compressor 0: Fan and compressor start simultaneously C: Fan continues to run (no	d	Effective when fan control mode F2=0. Note: When F2=1 or F2=2, this parameter item is disabled.

		delay in starting after dripping)		
		d: Fan continues to run, stops during defrosting and dripping		
		1 to 300: Fan starts 1 to 300 seconds after compressor		
F2	Fan controlled mode	0: Fan controlled by compressor time 1: Fan controlled by defrosting probe temperature 2: Fan controlled by cabinet temperature and defrosting probe temperature control	0	Fan operating mode, select one Note: When d1=0, this parameter item is disabled.
F3	Delay time for first fan start after defrosting and dripping Time	0 to 300 seconds	60 seconds	Delay time for the first fan start after defrosting and dripping (to prevent cabinet expansion)
F4	Fan start temperature	-40.0°C to fan stop temperature	-5.0°C	Effective under fan controlled mode F2=1 Note: When F2=0 or F2=2, this parameter item is disabled. Note: When d1=0, this parameter item is disabled.
F5	Fan stop temperature	Fan start-up temperature: ~50.0°C	10.0°C	Effective under fan controlled mode F2=1 Note: When F2=0 or F2=2, this parameter item is disabled. Note: When d1=0, this parameter item is disabled.

6.3 -60°C~-80°C models

6.3.1 Panel Schematic:



6.3.2 Work Instruction Status Description:

sign	instruction	statuses	mean
	Refrigeration Symbol	lights out	Uncooled state
		lights on	Refrigeration
		light flashing	Cooling state, stop cooling
	Parameter Setting Symbol	lights on	Parameter setting status
	warning symbol	lights on	fault state
	Standby Symbol	lights on	standby mode
	defrosting symbol	lights on	Frosting
	lock symbol	lights out	Stop Frosting
°C	Temperature Unit Symbols -Degrees Celsius	lights on	Locked status

6.3.3 Operating Instructions:

1、Standby function

(a) In the non-set state, press and hold the " " key for 3s to enter

Cabinet temperature over-limit alarm: A9=1, over-limit alarm mode selection is set value + over-temperature alarm deviation; when the cabinet temperature exceeds the temperature set value (St) + over-temperature alarm hysteresis (A8),

and the cabinet temperature over-limit alarm delay time has elapsed, the digital display shows rH. When the temperature is below the temperature set value (St) + over-temperature alarm hysteresis (A8), the alarm is released; When the cabinet temperature is less than the temperature setpoint (St) - over-temperature alarm hysteresis (A8), and the cabinet temperature over-limit alarm delay time has elapsed, the digital display shows rL. When the temperature exceeds the temperature setpoint (St) - over-temperature alarm hysteresis (A8), the alarm is cleared.

Condenser over-temperature alarm: cH displays the alarm without affecting system operation.

Condenser over-temperature shutdown alarm: cP displays the alarm, and the refrigeration system stops running. The system can resume operation half an hour after the alarm disappears.

F6	Fan start temperature difference value	0°C ~ +50.0°C	5.0°C	Fan control mode F2=2 is effective. Note: When F2=0 or F2=1, this parameter item is disabled. Note: When d1=0, this parameter item is disabled.
A1	Proportional output stop time	1 to 60 minutes	10 minutes	When the cabinet temperature sensor malfunctions, the compressor stops running.
A2	Proportional output start time	0 to 60 minutes	30 minutes	When the cabinet temperature sensor malfunctions, the compressor starts up.
A3	Enable alarm sound	0: No 1: Yes	1	Does the buzzer sound when an alarm is triggered?
A4	Cabinet temperature exceeds upper limit alarm value	Cabinet temperature exceeds lower limit alarm value ~ 85.0°C	40.0°C	Note: When A9=1, this parameter item is disabled.
A5	Cabinet temperature exceeds lower limit alarm value	-40.0°C to cabinet temperature upper limit alarm value	-30.0°C	Note: When A9=1, this parameter item is disabled.
A6	Cabinet temperature overheat alarm delay	0 to 120 minutes	20 minutes	An alarm will only be triggered if the temperature remains above the set point for longer than the time specified in this parameter.
A7	Delay in alarm for excessive cabinet temperature upon initial power-up	0 to 24 hours	2 hours	After power-on, no temperature overheat alarm signal occurs within the set time.
A8	Over-temperature alarm hysteresis	0 ~ 60°C	8.0°C	
A9	Over-temperature alarm	0: Absolute temperature	0	

	mode	point 1: Start/stop temperature ± Over-temperature alarm value		
A10	Integer and decimal display conversion	0: Decimal 1: Integer	0	
A11	Defrost relay reset function	0: Defrosting 1: Fan 2: Lighting 3: Standby	0	
d01	Door switch control output selection	0: Cancel door switch; 1: Turn off fan when door is open; 2: Turn on lights when door is open, turn off lights when door is closed; 3: Turn on lights when door is open, turn off fan; turn off lights when door is closed, fan returns to state before door was opened; 4: As a synchronized defrost signal, start defrosting; 5: Only as a door switch	0	
d02	Door opening buzzer response delay	0 to 200, 0: No response	0	
cd1	Condensation Probe Function Selection	0: Off 1: Turn on condensation probe	0	
cd2	Condenser high temperature alarm activation value	30℃ ~ cd4-1	55℃	

resolved, the system enters temperature control mode and exits the fixed on/off cycle mode.

When the defrost sensor malfunctions, the temperature display window alternately shows fault code E2 and the current cabinet temperature; the controller controls according to the set defrost cycle and defrost time. Once the defrost sensor malfunction is resolved, the system enters dual control based on time and temperature.

If both the cabinet temperature sensor and defrost sensor malfunction simultaneously, the temperature display window alternately shows fault codes E1 and E2;

Cabinet temperature overlimit alarm: A9=0, overlimit alarm mode is set to absolute temperature point; When the cabinet temperature exceeds the upper limit alarm value (A4) and the overlimit alarm delay time has elapsed, the digital display shows rH. The alarm is cleared when the cabinet temperature falls below the upper limit alarm value (A4). When the cabinet temperature falls below the lower limit alarm value (A5), and the cabinet temperature exceeds the lower limit alarm delay time, the digital display shows rL. When the cabinet temperature is greater than the lower limit alarm value (A5), the alarm is released.

c: The fan runs continuously.

d: The fan runs continuously and stops during defrosting and defrosting drainage.

1–300: The fan starts 1–300 seconds after the compressor.

F2=1: Fan controlled by defrosting probe temperature

F4: Fan start temperature

F5: Fan stop temperature

F2=2: Fan controlled by cabinet temperature and defrosting probe temperature difference

F6: Fan start temperature difference value; the fan starts when the temperature difference between the cabinet temperature and the defrosting probe temperature is greater than or equal to this value.

6.2.10 Alarm output:

When the cabinet temperature sensor malfunctions, the temperature display window shows error code E1; the controller controls the compressor to stop for 30 minutes and run for 15 minutes in a fixed pattern. After the cabinet temperature sensor malfunction is

cd3	Condenser high temperature alarm hysteresis	1°C ~ 15°C	5°C	
cd4	Condenser high temperature protection start value	cd2+1 ~ 90°C	64°C	
tEn-u1		Note: This interval parameter is reserved for future use.		
u2	Display delay for every 1°C increase in cabinet temperature	0 to 90 (unit: 10 seconds), 0: Cancel delay	3	
adr	mailing address	0-246	0	spare
pas	password	-127-127	55	

6.2.7 Refrigeration control:

Compressor start conditions: (all of the following conditions must be met)

The compressor shutdown time exceeds the set compressor delay time;

Cabinet temperature > temperature setpoint (St);

When the cabinet temperature is $\geq$ temperature setpoint (St) + refrigeration hysteresis or during hot gas defrosting, except during defrosting or defrosting drip conditions;

Compressor shutdown conditions (meet any of the following conditions)

When the cabinet temperature $\leq$ temperature setpoint (St);

When electric defrosting begins;

When hot gas defrosting ends.

6.2.8 Defrost control:

When the defrost cycle is not 0;

When the defrost sensor temperature is lower than the set defrost stop temperature;

When the set defrost cycle time has elapsed, or when forced defrosting begins.

Defrost end conditions: dual control of temperature and time. After defrosting starts, if the defrost probe temperature reaches the “defrost stop temperature,” defrosting is considered complete and ends. If the defrosting time is too long and exceeds the “defrosting time,” the controller will also end the defrosting process. If neither the temperature nor the time reaches the set value, the defrosting state can also be exited by manually pressing the “forced defrosting” button;

Defrosting drip time: The purpose of defrosting dripping is to drain the water accumulated on the evaporator during defrosting. If it cannot be drained in time, this water will freeze again during refrigeration.

Therefore, a defrost drip time must be set to ensure timely drainage of water accumulated on the evaporator during defrosting. During this period, the compressor will not start. For example, if the defrost drip time is set to 5 minutes, the compressor cannot start for cooling within 5 minutes after defrosting is completed.

6.2.9 Fan control:

The fan has three control modes (parameter F2):

0: The fan is controlled by the compressor time;

1: The fan is controlled by the defrost sensor temperature;

2: The fan is controlled by the temperature difference between the cabinet temperature and the defrost sensor temperature. These three control modes are mutually exclusive; only one function can be selected.

F2=0: The fan is controlled by the compressor time.

-180 to -1: The fan starts 180 to 1 seconds before the compressor.

0: The fan starts and stops simultaneously with the compressor.